



Recent advances in engineering microbial lipases for industrial applications

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ABSTRACT

Lipases, a green biocatalyst found in animals, plants, and microorganisms, are widely used in the agricultural, biofuel, cosmetics, chemical, food, pharmaceutical, and textile industries due to their versatility, exceptional specificity, and ease of production and use. However, the presence of multiple isoforms of microbial lipases often limits their applications and requires costly purification. There are also growing demands for improved lipase stability, activity, and specificity in industrial applications. One emerging research direction in this field is integrating synthetic biology and engineering tools to design novel lipases for diverse industrial applications. Recent progress in protein engineering, immobilization technologies, and artificial intelligence (AI) tools have significantly improved lipase catalytic performance. This paper provides a comprehensive review of the classification, general characteristics, industrial production and applications of lipases and recent advances in engineering lipases and lipase-producing microbial cells to develop novel lipase-based bioprocesses and bioproducts.

1. Introduction

Lipases are a group of hydrolase enzymes (triacylglycerol acyl hydrolases EC 3.1.1.3) widely found in animals, plants, and microorganisms that hydrolyze triglycerides into glycerol and free fatty acids (Gupta et al., 2004; Javed et al., 2018). Compared to animal and plant lipases, microbial lipases have wider industrial applications (Reetz, 2013) because they can be readily produced in large quantities at relatively low costs. Both naturally isolated and genetically engineered microbes can produce lipase at a high titer, productivity, and yield in fermentation with economical media. With exceptional stability in organic solvents, lipases can be used efficiently to catalyze reactions in aqueous as well as non-aqueous media (Tan et al., 2018). Lipases are widely used in the agricultural, biofuel, cosmetics, chemical, food, pharmaceutical, and textile industries to catalyze reactions including hydrolysis, esterification, transesterification (interesterification, alcoholysis, and acidolysis), and aminolysis, contributing to the multibillion-dollar lipid technology bioindustry (Hasan et al., 2006; Mahfoudhi et al., 2022). With increasing interest in industrial applications of lipases as

biocatalysts (Kempka et al., 2008; Vanleeuw et al., 2019), there are growing demands for lipases with enhanced properties like increased activity, enantio-selectivity, thermostability, reusability, and tolerance towards organic solvents. With advanced synthetic biology and computational tools (Gamboa-Melendez et al., 2018; Spartalis et al., 2024), customized lipases can be developed to specifically address emerging process requirements. Also, robust microbial strains can be developed using novel metabolic and genomic engineering strategies (Ali et al., 2023; Contesini et al., 2020) to improve the mass production of lipases for lipase-based industrial bioprocesses.

In this review, we first provide an overview of lipases for their uses in catalyzing various reactions for industrial applications and their sources, general characteristics, and production methods. Next, we discuss novel engineering strategies, including genetic engineering for developing highly efficient lipase producing microbes and protein engineering and enzyme immobilization for enhancing lipase catalytic performance for industrial applications. Finally, we highlight recent advances in applying artificial intelligence (AI) to lipase engineering and future prospects for the field.

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2. Lipase-catalyzed reactions and applications

Lipases are chemo selective (selectively catalyze one chemical reaction), regioselective (specifically acts on one substrate triglyceride), or stereoselective (selectively produce one of the possible stereoisomers) (Yao et al., 2021). The regioselective lipases can be further divided into three groups. First, non-specific lipases hydrolyze triglycerides into free fatty acids and glycerol (Utama et al., 2019); this group of lipases is highly promiscuous and demonstrates versatility in catalyzing various substrates. Second, 1,3-specific lipases release fatty acids from the 1 and 3 carbon positions of their triglyceride substrates (Gao et al., 2021). One example is biodiesel production using immobilized lipase from *Rhizopus oryzae*, which exhibits 1,3-specificity (Clementz et al., 2016). Third, fatty-acid-specific lipases hydrolyze fatty acids containing cis-double bonds between the 1 and 9 carbon positions (Kapoor and Gupta, 2012). Lipases also show selectivity on the chain length of the substrates. For example, the lipases derived from *A. niger*, *A. oryzae*, *Bacillus coagulans*, *Candida lipolytica*, *Geotrichum candidum*, and *Penicillium citrinum* exhibit strong specificity towards short-chain esters, while the lipases from *Mucor racemosus*, *Trichosporon capitatum*, and *R. oryzae* exhibit strong selectivity for medium- and long-chain as well as branched esters (Song et al., 2008).

Fig. 1 illustrates various types of reactions catalyzed by lipases with their notable applications listed in Table 1. Lipases can catalyze hydrolysis, esterification, aminolysis, and transesterification reactions, depending on the reaction conditions (Akoh et al., 2007). In general, lipase-catalyzed reactions rarely proceed in the direction of esterification in the presence of water, which promotes hydrolysis and inhibits esterification (Wang et al., 2021a). Most lipases display interfacial activation behavior and work more effectively in the presence of an organic solvent (Chandra et al., 2020). With versatile functions and wide industrial applications, lipases accounted for approximately 11.6 % of the global enzyme market valued at \$336 million in 2019, which increased to \$535 million in 2021 and is projected to reach \$990 million by 2031 (Bullo et al., 2024). Compared to chemical catalysis, biocatalysis with lipase has the advantages of lower energy consumption and decreased reliance on hazardous catalyst materials and waste generation. For example, the enzymatic hydrolysis of fat by lipases is preferred over traditional methods using elevated temperatures and

pressures that result in undesirable lipid oxidation (Salimon et al., 2011). Also, lipase-catalyzed esterification produces just water as a byproduct and does not use any hazardous solvents. Complete conversions can be achieved by in-situ water removal and a wide variety of alcohols (acceptor), and organic acids (donor) can be esterified by lipases, although different lipases with varied active pocket structures showed different selectivity of the substrate.

The economic feasibility of lipase applications varies with and depends on the processes used in lipase production, including fermentation, purification, and immobilization, and the lipase-catalyzed reactions, substrates, final products, and their intended applications or markets. It is easier to justify the financial costs of enzyme-based biomanufacturing for applications in the pharmaceutical industry, where lipase enables the production of high-value drugs. For example, Wang et al. (2023) reduced the production cost of brivaracetam by combining direct esterification catalyzed by porcine pancreatic lipase with a cyano-hydrolysis-cyclization cascade reaction, which was successfully scaled up to kilogram-level synthesis, significantly lowering the treatment cost for epilepsy. Kua et al. (2023) reported an innovative green chemistry approach to synthesize N-acyl glycines from fatty acids using an engineered lipase that significantly enhanced aminolysis activity and achieved high yields of various N-acyl glycines, which are commonly applied in cosmetic and pharmaceutical industries. Manzoor et al. (2024) utilized lipase-catalyzed interesterification to produce a trans-fat-free fat analogue from mustard oil and palm stearin, resulting in a healthier alternative to commercial shortening with similar functional properties but improved nutritional benefits. The purity of lipase is important in the pharmaceutical and cosmetic industries as enzyme purification enhances its applications (Gullon et al., 2017). Lipases produced in fermentation are usually recovered and separated via ammonium sulfate precipitation followed with several chromatographic steps for further purification based on the lipase sources and their protein size (Saxena et al., 2003). Baradia et al. (2023) utilized SuperPro Designer to simulate the production and purification of lipase (laboratory-grade enzyme) from *Bacillus subtilis*. Despite a recovery rate of only 50 %, the process achieved a payback period of approximately 1.37 years. Khanra et al. (2025) provide a more in-depth discussion of the lipase-mediated sustainable synthesis of pharmaceuticals and chiral organic building blocks. The application of lipase in esters production is also receiving more attention because of its attractive economics. Faisal et al. (2024) reported that the synthesis of 2-ethylhexyl oleate, primarily used as a non-toxic plasticizer and lubricant, using Novozym 435 resulted in a production cost of \$2109 per ton, yielding a 28 % profit margin compared to prevailing market prices.

For lipase applications in the production of chemicals and biofuels, the enzyme cost associated with lipase production processes could be a major limiting factor. It should be noted that lipase production levels varied greatly with the producing microorganisms as well as the process conditions (see Tables 2, 3). Solid-state fermentation (SSF) and submerged fermentation (SmF) are two well-established techniques for enzyme production. Almost 90 % of the industrial enzymes are produced by SmF due to the high-water activity (>0.9) that bacteria normally require to grow compared to SSF (Treichel et al., 2009). SmF is thus widely used for microbial lipase production, often resulting in higher thermostability (Colla et al., 2010; Colla et al., 2015; Dutra et al., 2008). However, lipases from filamentous fungi were often produced in SSF because they could grow extensively both inside and outside of solid substrates, allowing for intimate contact with the substrate and thereby enhancing enzyme production. A techno-economic analysis comparing SmF and SSF for lipase production in *Penicillium restrictum* showed that SmF was economically unfeasible, as the production cost exceeded the market price (Castilho et al., 2000). In contrast, SSF significantly reduced costs, yielding a unit cost 47 % lower than the retail price and a payback period of 1.5 years. More recently, de Barros et al. (2025) reported lipase production via SSF using distiller's dried grains, achieving a production cost of \$3.47/kg and a profit margin exceeding 100 %, with

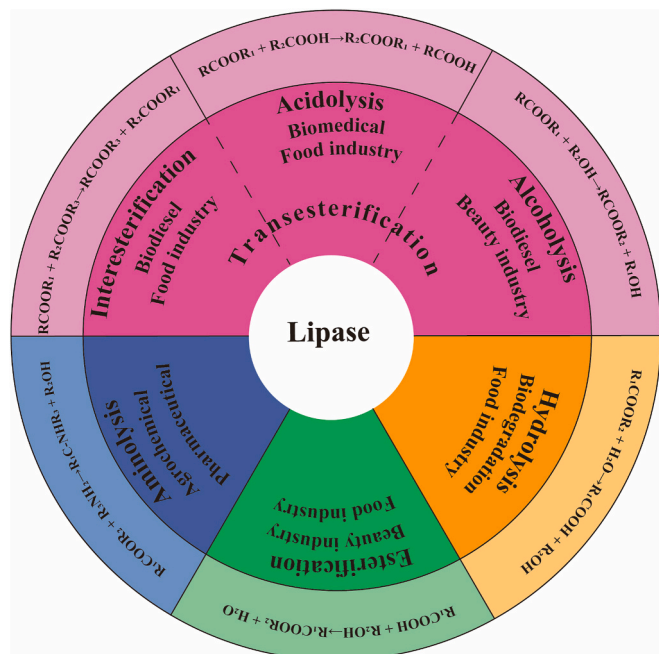


Fig. 1. Reactions catalyzed by lipases and their applications.

Table 1
Examples of lipase-catalyzed reactions and applications.

	Substrate	Lipase source	Application	Reference
Hydrolysis	Sunflower oil	<i>C. rugosa</i>	Fatty acids synthesis	Kim and Hou, 2006
	Long chain triacylglycerols	<i>B. pumilus</i>	Food and detergent industry	Laachari et al., 2015
	Phorbol 12-myristate 13-acetate	<i>P. aeruginosa</i>	Solid waste treatment	Bose and Keharia, 2013
	Edible oils	<i>S. marcescens</i>	Detergent	Zaki and Saeed, 2012
	Soybean	<i>A. oryzae</i>	Production of miso	Abe et al., 2006
	Olive oil	<i>C. antarctica</i>	Fatty acids synthesis	Santos et al., 2015
	Soybean oil	<i>Y. lipolytica</i>	Fatty acids synthesis	Huang et al., 2023
	Effluents from fast restaurants	<i>S. capitis</i>	Wastewater treatment	Rmili et al., 2024
	Fatty acids and alcohols	<i>C. antarctica</i> , <i>R. miehei</i> , <i>T. lanuginosa</i>	Flavor esters	Martins et al., 2014
	Jatropha oil	<i>R. miehei</i>	Oil refining	Goffert et al., 2014
Esterification	Alcohol and acid	<i>Geotrichum</i> sp.	Food and beauty industry	Dhake et al., 2013
	Glycerol with palmitic acid	<i>C. antarctica</i>	Food, soaps, cosmetics	Kapoor and Gupta, 2012
	Fatty acid and alcohols	<i>S. epidermidis</i>	Food industry	Patil et al., 2011
	Fatty acids and alcohol	Porcine pancreatic	Flavor esters	Gonçalves et al., 2021
	Mangiferin and fatty acids	<i>T. lanuginosus</i>	Food industry	Yu et al., 2024
	Linoleyl ethyl ester and aminoalcohol	<i>C. antarctica</i>	Cutaneous application	Couturier et al., 2009
	Racemic amines and ethyl acetate	<i>C. antarctica</i>	Pharmaceutical and agrochemical intermediate	Xu et al., 2015
Aminolysis	Esters or carboxylic acids and amines	<i>Sphingomonas</i> sp. HXN-200	Pharmaceutical intermediate	Zeng et al., 2018
	Fatty acids and glycine	<i>R. miehei</i>	Cosmetic and pharmaceutical industries	Kua et al., 2023
	Ester and amine	<i>R. dicambivorans</i>	Pharmaceutical industry	Wang et al., 2024a
	Biodiesel	<i>P. camembertii</i>	Biodiesel modification	Padhi et al., 2012
	Water polluted by phthalates	<i>C. rugosa</i>	Biodegradation	Mita et al., 2010
	Mustard oil and palm stearin	<i>T. lanuginosus</i>	Food industry	Manzoor et al., 2024
	Trimethylolpropane	<i>C. antarctica</i>	Biomedical	Sayed et al., 2015
Transesterification	Triolein, olive oil and fatty acid	<i>Y. lipolytica</i>	Biomedical	Akil et al., 2020
	Triglyceride and fatty acids	<i>R. japonicus</i>	Food industry	Yoshioka et al., 2023
	Soybean oil and methanol	<i>A. oryzae</i>	Biodiesel production	Huang et al., 2022
	Licuri oil and grain alcohol	<i>C. antarctica</i>	Beauty industry	Moreira et al., 2024

a payback time of 1.46 years. These studies underscore that the outcomes of economics heavily rely on lipase costs and its intended market and applications.

Biodiesel production catalyzed by lipase is also highly influenced by the choice of substrates. Budzaki et al. (2019) utilized waste cooking oil as a low-cost feedstock and further reduced costs by immobilizing lipase on oil cake. Despite these measures, the cost of lipase remained a major limitation in achieving an environmentally sustainable biodiesel production process. Using commercial lipase, a positive net present value was achieved after one year of producing 10,145.91 tones of biodiesel. In contrast, lipase produced via SSF was found to be economically unfeasible in their biodiesel production scenario. Collectively, the costs of lipase and substrates are the key factors influencing the economic viability of its applications. A one-pot process that integrates lipase production with the catalytic reaction would offer a more attractive and cost-effective solution.

In summary, it is important to develop cost-effective and highly productive and selective lipases for industrial applications. How to engineer microbes for enhanced lipase production and to engineer lipase with enhanced activity, selectivity and stability are thus of high interest and are discussed in detail in Sections 3 and 4, respectively. Furthermore, immobilization can enhance lipase recyclability, stability, activity, and reduce lipase cost in industrial applications. Methods for lipase immobilization and their benefits for lipase-based bioprocesses are discussed in Section 5.

3. Engineering microbes for enhanced lipase production

Table 2 lists notable microbial lipases, their production hosts, substrates, and activity levels achieved in fermentation. Fungal lipases have been extensively researched since the 1950s because of their remarkable thermal and pH stability, substrate specificity, and ability to function in organic solvents (Mehta et al., 2017). These unique characteristics have made fungal lipases highly desirable for various applications. Among them, *Candida antarctica* lipase B and *Candida rugosa* lipase are the two most popular industrial lipases since these yeasts are “Generally Recognized as Safe” (GRAS) (De Maria et al., 2007). Important lipase-

producing filamentous fungi include *Rhizopus* sp., *Aspergillus* sp., and *Penicillium* sp. *Bacillus* lipases are the most common bacterial lipases (Treichel et al., 2009). *Pseudomonas* sp., *Staphylococcus* sp., *Arthrobacter* sp., and *Chromobacterium* sp. were also used for lipase production (Sardesai and Bhosle, 2004; Bharathi and Rajalakshmi, 2019). Microbial lipases exhibit different cellular localizations, including extracellular, intracellular, and membrane bound. Their high levels of activity, stability, and wide variety of hydrolytic reactions enable them to fit various industrial processes (Javed et al., 2018; Panizza et al., 2013).

Besides using native microbial hosts to produce lipase, extensive research has been done on expressing lipase genes in a variety of heterologous hosts at a much higher expression level and lipase activity (Table 3). *E. coli* is the most widely used bacterial host (Contesini et al., 2020) and *Pichia* is the most used yeast host because of their well-established genetic engineering tools, relatively easy mass cultivation, and high enzyme production efficiency. Various engineering strategies including codon optimization, promoter optimization, facilitated protein folding with chaperons, facilitated transport with signal peptides, and surface display have been applied to enhance lipase production in heterologous hosts (Fig. 2). Some notable studies are listed in Table 4 with detailed discussion given in the following sections.

3.1. Codon optimization for protein expression

Codon usage is the key factor that influences the efficiency of heterologous gene expression. In addition to differences in codon frequencies, some microorganisms exhibit non-standard codon decoding. For example, CTG usually encodes leucine but is read as serine in *C. rugosa* (Kawaguchi et al., 1989). Consequently, almost no active lipase can be obtained by directly cloning the *C. rugosa* lipase gene in a heterologous host such as *Pichia pastoris*. Therefore, codon optimization is often used to improve heterologous lipase production. To overcome non-standard codons (i.e., CTG encoding serine instead of leucine), Brocca et al. (2000) first produced four mutant lipase genes with 2 to 8 CTGs replaced with TCT (the standard codon for serine); however, only intracellularly inactive lipase was produced at 10–20 mg/L in *Pichia pastoris*. On the other hand, total synthesis of lipase with substitution of

Table 2

Lipases produced by native microbial species in fermentation.

Host	Inducer	Substrate*	Lipase Activity, Fermentation time	References
Bacteria				
<i>Acinetobacter</i> sp. AU07	Castor oil	Medium	48 U/mL, 16 h	Gururaj et al., 2016
<i>Acinetobacter</i> sp.	Coconut oil	Coconut oil cake	32.2 U/mg protein, 24 h	Khoramnia et al., 2011
<i>A. radioresistens</i>	None	Tween 80	132 U/mL, 15 h	Li et al., 2005
<i>Aeromonas</i> sp. S1	None	De-oiled seed cake	195 U/mL, 24 h	Mahdi et al., 2012
<i>B. coagulans</i>	None	Medium	80 U/mL, 72 h	Daouadji et al., 2015
<i>B. subtilis</i>	Olive oil		4.96 U/mL, 84 h	Suci et al., 2018
<i>Burkholderia</i> sp.	None		122.3 U/mL, 36 h	Lo et al., 2012
<i>B. stabilis</i>			25 U/mL, 72 h	Yoshida et al., 2019
<i>G. thermodenitrificans</i>			330 U/mL, 15 h	Christopher et al., 2015
<i>G. thermoleovorans</i>			>1000 U/mL, 18 h	Abol Fotouh et al., 2016
<i>Staphylococcus warneri</i>	Olive oil		17.21 U/mL, 48 h	Yele and Desai, 2015
<i>T. lanuginosus</i>	Olive oil, triacetin		38.73 U/mL, 9 days	Sreelatha et al., 2017
Yeasts				
<i>C. cylindracea</i>	None	Oleic acid	23.7 U/mL, 170 h	Kim and Hou, 2006
<i>C. rugosa</i>	None	Soybean molasses	12.3 U/mL, 12 h	de Moraes et al., 2016
		Lactic, oleic acids	27 U/mL, 24 h	Montesinos et al., 2003
		Oleic acid	55 U/mL, 24 h	Gordillo et al., 1998
	Olive oil	Coconut oil cake, wheat bran	118.2 U/g, 72 h	Benjamin and Pandey, 1998
<i>C. tropicalis</i>	Olive oil	Olive mill waste	10.7 U/mL, 72 h	Kebabci and Cihangir, 2022
<i>Candida</i> W3.8	None	Medium	2357 U/mL, 120 h	Alami et al., 2017
<i>Cryptococcus</i> sp. MTCC 5453	None	Cottonseed oil cake	753 U/g, 120 h	Thirunavukarasu et al., 2016
<i>M. guilliermondii</i>	None	Medium	285.8 U/mL, 24 h	Knob et al., 2020
<i>R. mucilaginosa</i>	Olive oil		72 U/mL, 96 h	Potumarthi et al., 2008
<i>Y. lipolytica</i>	Olive oil		13 U/mL, 60 h	Gonçalves et al., 2013
			340 U/mL, 72 h	Darvishi and Hosseini, 2015
	Soy oil, Tween 80	Andiroba oil cake, soybean meal	63.7 U/g, 24 h	Carvalho et al., 2023
	None	Medium	11,000 U/mL, 136 h	Pan et al., 2023
Filamentous fungi				
<i>A. aculeatus</i>	None	Medium	9.5 U/mL, 214.7 h	Roy et al., 2021
<i>A. flavus</i>	Soybean oil		4 U/mL, 96 h	Colla et al., 2016
<i>A. niger</i>	None	Rice husk, cottonseed cake, red gram husk	28.2 U/g substrate, 96 h	Nema et al., 2019
		Wheat bran, gingelly oil cake	384 U/g substrate, 72 h	Mala et al., 2007
	Soybean oil	Medium	2171 U/mL, 48 h	Jia et al., 2015
<i>Aspergillus</i> sp.	Olive oil	Wheat bran	1934 U/g, 96 h	Adinarayana et al., 2004
<i>A. terreus</i>	Dextrose	Mustard oil cake	525–1366 U/mL, 96 h	Sethi et al., 2012, 2016
<i>P. citrinum</i>	Olive oil	Medium	123.2 U/g support, 120 h	Lima et al., 2017
<i>P. roqueforti</i>	Palm oil	Coco bran	33.3 U/g dry bran, 72 h	Araujo et al., 2022
<i>P. verrucosum</i>	None	Soybean bran	52 U/g dry bran, 72 h	Kempka et al., 2008
<i>R. chinensis</i>	Olive oil	Maltose, peptone	13.9 U/mL, 72 h	Teng and Xu, 2008
<i>R. oryzae</i>	Olive oil	Wheat bran	62.7 U/mL, 84 h	Hermansyah et al., 2019
		Medium	1.46 g cell/g support	He et al., 2016
			12 U/mL, 72 h	Reis et al., 2022
<i>T. harzianum</i>	Olive oil	Castor oil cake, sugarcane	4 U/g	Coradi et al., 2013

* Unless noted, synthetic media containing glucose as the carbon source and various nitrogen sources are used as the fermentation substrate.

all CTG codons resulted in the expression of active lipase at 150 activity unit per mL medium. Codon optimization in *Pichia pastoris* led to the production of *C. rugosa* lipase at 1200 U/mL (Xu et al., 2016). Similarly, *Pseudomonas plecoglossicida* S7 lipase production in *E. coli* increased to 2388.9 U/mg after codon optimization (Choudhary et al., 2023). Furthermore, the lip3 gene was codon-optimized to increase lipase production by 50–70 times (Chang et al., 2006).

PCR and gene synthesis can be used to express active lipases (with site mutagenesis) in heterologous hosts. Xu et al. (2010) cloned the *C. rugosa* lipase with codon optimization through overlap extension PCR to replace all 17 CTG codons with the standard TCT codon for serine and expressed the recombinant gene in *Pichia pastoris*, which produced active lipase. Further systematic codon-optimization to modulate codon adaptation index, GC content, and stem-loop structures increased the expressed lipase activity to 1200 U/mL (Xu et al., 2016).

3.2. Promoter optimization

Another strategy used to significantly boost microbial lipase production is promoter optimization. Potent constitutive promoters like XPR2, TEF, and RPS7 as well as induced promoters like ICL1, POT1, and POX2 have been created for the large-scale production of lipase from *Y. lipolytica* (Madzak et al., 2004). A high-level production of *Y. lipolytica*

lipase at 38,500 U/mL in *Pichia pastoris* was achieved with codon optimization and promoter and expression system selection. Vidal et al. (2023) reported that the *Rhizopus oryzae* lipase activity increased two-fold when driven by hybrid erythritol-inducible promoter pEYL1-5AB. Fusing a promoter with an upstream activating sequence (UAS) is also a powerful strategy to enhance lipase expression. Jiao et al. (2022) demonstrated that promoter upstream fusion of multi-copies of UAS increased *Thermomyces lanuginosus* lipase production in *Y. lipolytica* by 14.9-fold.

3.3. Molecular chaperones for protein folding

In addition to codon optimization, which affects protein translation, protein folding and transportation could also influence expression and final enzyme activity. It is well known that inclusion bodies of inactive or insoluble protein accumulate intracellularly because of heterologous protein production in *E. coli*. Many periplasmic folding factors, including DegP, FkpA, DsbA, and DsbC, were co-expressed with lipase B from *Pseudozyma antarctica* in *E. coli* (Xu et al., 2008). Lipase production was significantly improved when co-expressing chaperones including endoplasmic reticulum oxidoreduction 1 (ERO1) and immunoglobulin heavy chain binding protein (BiP) (Xu et al., 2016). With codon optimization and co-expression of molecular chaperones, *C. rugosa* lipase can be

Table 3

Lipases produced in heterologous hosts in fermentation.

Host	Lipase gene source	Inducer	Lipase Activity, Fermentation time	References
Bacteria				
<i>B. subtilis</i>	<i>G. stearothermophilus</i>	IPTG	5.6 U/mL, 48 h	Elemosho et al., 2021
	<i>P. fluorescens</i> lipB		78.2 U/mg, 6 h	Hu et al., 2023
<i>B. subtilis</i> WB-L01	<i>B. subtilis</i> lipase	None	57.5 U/mL, 72 h	Miao et al., 2022
<i>C. glutamicum</i>	<i>C. antarctica</i>		140 U/mL, 72 h	Gonzalez et al., 2019
<i>E. coli</i>	<i>Thermococcus onnurineus</i>		400 U/mg, 24 h	Soni et al., 2019
<i>E. coli</i> BL21	<i>Micrococcus luteus</i> EMP48-D		15.2 U/mg, 24 h	Nur et al., 2023
	<i>P. aeruginosa</i>		6595.7 U/mg, 6 h	Ganesh et al., 2025
	<i>B. licheniformis</i> NCU CS-5		161.7 U/mL, 12 h	Zhao et al., 2022
	<i>B. subtilis</i>		16.7 U/mL, 48 h	Gao et al., 2025
	<i>Acinetobacter haemolyticus</i>		292.5 U/mL, 16 h	Batumalaie et al., 2018
	<i>Burkholderia contaminans</i>		127 U/mL, 16 h	Alnoch et al., 2018
	<i>C. acetobutylicum</i>		400 U/mg, 18 h	Nagaroor and Gummadi, 2020
<i>E. coli</i> BL21(DE3)	<i>Drosophila melanogaster</i>	IPTG	NA	Alfaro-Chávez et al., 2019
	<i>Glaciecola nitratireducens</i>		4.2 U/mg, 16 h	Li et al., 2020
	<i>Proteus</i> sp. NH 2-2	Lactose	1.5 U/mL, 10 h	Shao et al., 2019
	<i>Pseudomonas</i> sp. LSK25		100 U/mL, 16 h	Salwoom et al., 2019
	<i>Streptomyces scabiei</i> 87.22	IPTG	1800 U/mg, 2-12 h	Greicius et al., 2023
	<i>T. lanuginosus</i>		1300 U/mg, 36 h	Beklemishev et al., 2022
<i>E. coli</i> BL21(DE3) pLysS	<i>Burkholderia terrortii</i> GP3	IPTG	16.5 U/mL, 16 h	Putra et al., 2019
	<i>Geobacillus zalihae</i> HT1		19.1 U/mg, 12 h	Ishak et al., 2019
<i>E. coli</i> Lemo21(DE3)	<i>Aquabacterium parvum</i> lipase	IPTG	162 U/mL, 24 h	Ho et al., 2023
<i>E. coli</i> BL21-Star (DE3)	<i>B. amyloliquefaciens</i>		113 U/mg	Khan et al., 2020
<i>E. coli</i> Origami B (DE3)	<i>Geobacillus zalihae</i> T1 lipase	IPTG	722.4 U/mL, 6 h	Hamdan et al., 2023
<i>E. coli</i> (pET SUMO)	<i>Pseudomonas plecoglossicida</i> S7		2388.9 U/mg, 1 h	Choudhary et al., 2023
<i>E. coli</i> TOP10	<i>Aneurinibacillus thermoaerophilus</i>	Arabinose	72.5 U/mL, 8 h	Masomian et al., 2018
<i>P. aeruginosa</i> SDK-6	<i>B. subtilis</i>		50.5 IU/mL, 72 h	Kaur et al., 2024b
Yeast				
	<i>C. antarctica</i>	Methanol	444.5 U/mL, 216 h	Xiao et al., 2023
	<i>C. rugosa</i> Lip1	Methanol	95 U/mL, 24 h; 13,490 U/mL, 130 h	Li et al., 2016; Li et al., 2018
<i>P. pastoris</i>	<i>T. lanuginosus</i>	Oleate	8700 U/mL, 60 h	Xu et al., 2023
	<i>Y. lipolytica</i>	Methanol	7770 U/mL, 60 h	
	<i>Y. lipolytica</i>	Methanol	12,500 U/mL, 180 h; 38,500 U/mL, 126 h	Yu et al., 2007; Zhou et al., 2015
<i>P. pastoris</i> GS115	<i>Proteus vulgaris</i> K80	Methanol	117 U/mL, 168 h	Yan et al., 2025
	<i>Rasamsonia emersonii</i> LIPR		13,620 U/mL, 168 h	Wang et al., 2024b
<i>P. pastoris</i> X33	<i>T. lanuginosus</i>	Methanol	6100 U/mg, 4 days	Beklemishev et al., 2022
	<i>Y. lipolytica</i> Lip2		220.9 U/mL, 72 h	Mohanty et al., 2024
<i>S. cerevisiae</i>	<i>K. marxianus</i> L-2029	Galactose	1.36 U/mg, 16 h	Martínez-Corona et al., 2024
<i>T. reesei</i>	<i>C. antarctica</i>	lactose	44 U/mL, 96 h	Li et al., 2024b
	<i>C. antarctica</i> LipB	None	11,000 U/mL, 136 h	Pan et al., 2023
<i>Y. lipolytica</i>	<i>R. oryzae</i>	None	266.7 U/mg, 72 h	Vidal et al., 2023
	<i>T. lanuginosus</i>	None	27 U/mg, 24 h; 1197 U/mL, 120 h	Jiao et al., 2022

Note: Synthetic media with chemical-grade carbon and nitrogen sources were used in the fermentation.

produced in *Pichia pastoris* at the highest level of 13,490 U/mL (Li et al., 2016). Similarly, co-expression of the chaperone *N*-acetyltransferase Mpr1 significantly improved secretion and stability of *Proteus vulgaris* K80 lipase in *Pichia pastoris*, reaching 117 U/mL (Yan et al., 2025).

3.4. Transporter for protein secretion

Another critical step for lipase production is to enhance its secretion or transportation across the cell membrane. The ABC protein transporter is widely used to secrete various extracellular products in Gram-negative bacteria (Green and Mecsas, 2016). *E. coli* can release a thermostable lipase (TliA) more easily with a designed ABC transporter made up of TliD, TliE, and TliF (Eom et al., 2005). TliA lipase production increased nearly three-fold when TliA and mutant TliD were co-expressed, although the expression levels of these transporter proteins remained the same. This discovery demonstrated that engineered transporter proteins can increase lipase secretion. Fang et al. (2020) also reported that the MinD C-tag facilitated the secretion of alcohol acyltransferase ATF1. This strategy could also be used for lipase transportation. Most lipases have signal peptides, which are vital for their secretion. A 23 % increase in extracellular lipase activity was obtained by using the SP6 signal peptide instead of the wild-type signal peptide (Vidal et al., 2023). A six-fold increase in lipase activity was achieved by replacing the native signal sequence with a more robust one derived from the lipase A gene of *B. subtilis* (Elemosho et al., 2021).

3.5. Surface display

Cell surface display is an effective method for expressing target proteins by fusion on anchoring motifs displayed on the host cell's surface (Chen and Georgiou, 2002). Baek et al. (2010) created a cell surface display system using the anchoring motif of OmpC to express *Pseudomonas fluorescens* lipase TliA in *E. coli* and obtained >10-fold higher lipase activity than those obtained with other motifs such as OprF and FadL. Surface display can also simplify the lipase recovery process and facilitate enzyme immobilization and application. Lu et al. (2023a) employed the carboxyl terminal domain of the protein encoded by *estA* from *P. aeruginosa* as the carrier to display the lipase on the membrane of *E. coli*, which improved extracellular transport and enzyme stability. The same anchor was fused to the lipase from *C. acetobutylicum* for surface display on *E. coli*, which exhibited 85 % of the activity of the commercial lipase Novozym 435 in butyl butyrate production (Lu et al., 2023b). Applying cell surface display in Gram-negative bacteria is generally easier than in Gram-positive bacteria due to their simpler cell envelope structure. However, the differences in outer layer structures among various hosts should be carefully considered when selecting an appropriate anchor protein.

3.6. Tools for strain engineering

Nowadays, CRISPR-Cas9 mediated genome editing is widely used to

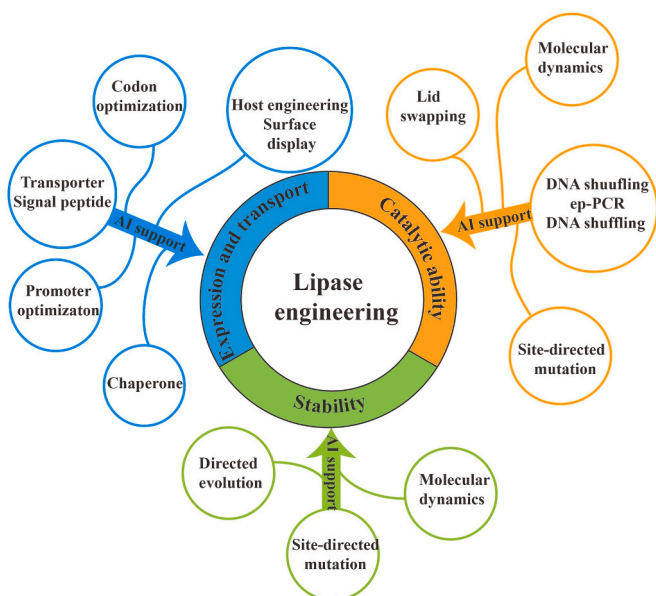


Fig. 2. Engineering strategies for improving production and performance of microbial lipases. Genetic engineering techniques including codon optimization, promoter optimization, chaperone-aided protein folding, signal peptide engineering for protein transportation and secretion, and host engineering for surface display are used to increase lipase expression and production, while protein engineering is used to improve lipase catalytic activity, selectivity, and stability, employing advanced synthetic biology, molecular dynamics simulation, and AI tools.

enable precise control over lipase gene expression, improving yields and stability (Shi et al., 2021). Dynamic regulatory gene circuits, employing biosensors and feedback loops, can be used to optimize enzyme production in real-time, reducing metabolic burdens on host cells (Boada et al., 2022). Systems biology tools including transcriptomics, proteomics, and metabolomics enable a deeper understanding of host metabolism, thereby facilitating targeted genetic modifications (Xu et al., 2020). For example, artificial intelligence (AI) can be used to identify lipase-producing yeasts through the analysis of gene expression data. Sundaramahalingam et al. (2024) trained an artificial neural network (ANN) using transcriptomic data and achieved an accuracy of 87.77 % in classifying the lipase-producing capability of yeasts. However, employing AI to engineer host strains for improved lipase yield is still in its infancy. Furthermore, AI, especially machine learning (ML), is becoming increasingly prevalent in systems metabolic engineering (Kim et al., 2020). To date, all studies involving AI-assisted lipase production have primarily focused on optimizing culture conditions to enhance yield (Ge et al., 2023). For instance, Amenaghawon et al. (2024) employed six machine learning models to predict lipase production by *Aspergillus niger* using a formulated waste-derived substrate. Among these models, support vector regression (SVR) demonstrated the most accurate predictive performance. Further optimization using manta ray foraging optimization (MRFO), particle swarm optimization (PSO), genetic algorithm, and surfactant induction led to a 168.3 % increase in the lipase yield (520.95 U/g dry substrate). The integration of ML models for culture optimization can enhance predictive capabilities and streamline the strain engineering process (Sarmah et al., 2022).

In summary, advanced synthetic biology tools will pave the way for the next generation of industrial biocatalysts with superior efficiency and sustainability. The continuous refinement of codon optimization, chaperone co-expression, secretion signal engineering, and promoter optimization, combined with novel host engineering approaches, will drive significant advancements in the production of microbial lipases in heterologous systems. The integration of multi-omics data will further promote the application of ML in metabolic pathway reconstruction and

metabolic flux optimization. The future of microbial cell engineering for lipase production will center on integrating multi-omics strategies with AI-driven tools to enhance precision and efficiency.

4. Lipase engineering

Most lipases have a molecular weight between 19 and 60 kDa (Sharma and Kanwar, 2014). Lipases usually contain an N-terminal signal peptide, which mediates transportation and secretion. Lipases are active and stable in non-aqueous media due to the presence of disulfide bridges (Priyanka et al., 2019). The activity and performance of lipases are highly dependent on the nature of their substrates, including carbon-chain length, unsaturation, and position of the fatty acid chain (Carpen et al., 2019). In general, cofactors are not required for lipase activity. However, the presence of metals and chemical inhibitors may affect lipase activities. For example, Co^{2+} , Ni^{2+} , ethylenediaminetetraacetic acid (EDTA), and sodium dodecyl sulfate (SDS) significantly reduce lipase activity, whereas Ca^{2+} increases it (Castro et al., 2016). In addition, lipases exhibit activity that is dependent on the pH. Fungal lipases can be classified into three subclasses (GX, GGGX, and Y) based on the presence of the oxyanion hole and their amino acid sequence (Kumar et al., 2023; Mehta et al., 2017). The GX class consists of lipases found in filamentous fungi and *C. antarctica* lipase B, the GGGX class includes lipases from *C. rugosa* and *Y. lipolytica*, while *C. antarctica* lipase A is categorized under the Y class.

In general, natural lipases must be optimized to be suitable for specific industrial applications (Fjerbaek et al., 2009). The optimal temperature range for natural lipases is between 30 and 50 °C, at which the transesterification reaction proceeds slowly, making the process time-consuming and less competitive in the market. Hence, improving the thermostability of lipase is essential for the reactions to occur at a higher temperature (>50 °C) and reaction rate. Furthermore, in the presence of solvents, natural lipases usually lose their catalytic activities over time and have limited lifetimes and must be replaced frequently in industrial reactors. A lipase will have a longer lifetime if it is resistant to solvents and its reaction products (Lu et al., 2012). Finally, while industrial processes prefer adaptability to a variety of feedstocks with varying compositions, most natural lipases have been developed to specifically act on substrates with predetermined chain lengths. To be commercially viable, lipase must be engineered with altered or enhanced substrate capabilities (Hwang et al., 2014). Site-directed mutagenesis is a useful tool for providing insight into lipase catalysis and specificity. The biocatalytic properties of an enzyme are related to its 3-dimensional structure or conformation. Thus, rational design based on protein conformation modifications can be used in protein engineering to improve its catalytic performance (Akoh et al., 2007). In addition, since enzyme immobilization can often increase lipase performance (productivity and stability), it is also necessary to engineer the lipase to facilitate its immobilization on targeted support materials. The strategies to engineer lipases for industrial applications are illustrated in Fig. 2 with approaches shown in Fig. 3. Notable studies on engineering lipases for enhanced enzyme catalytic ability and stability are summarized in Table 5 and discussed below.

4.1. Improved catalytic activity

For most lipases, the active sites are covered by a lid that undergoes conformational changes to allow substrate entry (Fig. 4A). The α -helical loop (lid) covers the catalytic domain and undergoes a conformational change that allows the substrate access to the active site (Khan et al., 2017). The lipase-mediated reaction starts with an acylation. Then the serine in the active site attacks the carbonyl group of substrates to form a tetrahedral intermediate. Consequently, water or alcohol can attack the intermediate (Godoy et al., 2013). Since the lid is the substrate-interacting site, it influences the enzyme activity and specificity (Chen et al., 2020). Thus, lipase catalytic ability and substrate specificity can

Table 4

Genetic engineering strategies employed to enhance lipase production in microorganisms.

	Method	Lipase gene source and heterologous host	Outcome	Reference
Increasing Expression Efficiency	Codon optimization	<i>C. rugosa</i> Lip3 in <i>P. pastoris</i>	Lipase increased by 50–70 times	Chang et al., 2006
		<i>C. rugosa</i> Lip1 in <i>P. pastoris</i>	1200 U/mL in <i>P. pastoris</i> cells	Xu et al., 2016
		<i>Geobacillus zalihae</i> T1 lipase in <i>E. coli</i>	722.43 U/mL	Hamdan et al., 2023
		<i>P. aeruginosa</i> lipase in <i>E. coli</i>	Specific activity of 6595.71 U/mg	Ganesh et al., 2025
		<i>T. lanuginosus</i> lipase in <i>E. coli</i>	Specific activity of 1300 U/mg	Beklemishev et al., 2022
	Overexpression	<i>T. lanuginosus</i> lipase in <i>P. pastoris</i>	6100 U/mg in <i>P. pastoris</i> cells	Beklemishev et al., 2022
		<i>E. coli</i> lip886 and lip385 in <i>Saccharopolyspora spinosa</i>	5.5-fold increase in spinosad production	Cao et al., 2024
	Promoter optimization	<i>B. subtilis</i> lipase in <i>B. subtilis</i> WB800N	Lipase production at 57.5 U/mL	Miao et al., 2022
	Stronger promoter	<i>R. oryzae</i> lipase in <i>Y. lipolytica</i>	Two times higher lipase activity	Vidal et al., 2023
	Lactose-induced promoter	<i>S. scabiei</i> lipase in <i>E. coli</i>	1800 U/mg for GD-95RM lipase	Greicius et al., 2023
	Promoter upstream fusion of multi-copies of UAS	<i>T. lanuginosus</i> lipase in <i>Y. lipolytica</i>	Lipase activity increased by 14.9-fold	Jiao et al., 2022
	Promoter optimization and signal peptide engineering,	<i>Y. lipolytica</i> Lip2 in <i>P. pastoris</i>	Lipase activity of 220.89 U/mL	Mohanty et al., 2024
	Expressing chaperones ERO1p, BiP	<i>C. rugosa</i> Lip1 in <i>P. pastoris</i>	Lipase production at 1758 U/mL	Xu et al., 2016
	Codon optimization and co-expression of molecular chaperones	<i>C. rugosa</i> lipase in <i>P. pastoris</i>	Lipase production at 13,490 U/mL	Li et al., 2016
Protein Folding	Co-expression of the chaperone <i>N</i> -acetyltransferase Mpr1	<i>Proteus vulgaris</i> K80 lipase in <i>P. pastoris</i>	Improved secretion and stability, reaching 117 U/mL	Yan et al., 2025
	Chaperone expression, MSS signal peptide	<i>Rasamsonia emersonii</i> LIPR in <i>P. pastoris</i>	Enzyme activity of 13,620 U/mL	Wang et al., 2024b
	Fusion with the mutated transporter	<i>P. fluorescens</i> lipase in <i>E. coli</i>	Around 3-fold higher secretion	Eom et al., 2005
	More robust signal sequences	<i>R. oryzae</i> lipase in <i>Y. lipolytica</i>	23 % improvement in lipase activity	Vidal et al., 2023
		<i>G. stearothermophilus</i> lipase in <i>B. subtilis</i>	Lipase activity increased by 6-fold	Elemosho et al., 2021
	Secretion and transport enhancement	<i>P. fluorescens</i> Lipase B in <i>B. subtilis</i>	Specific activity of 78.17 U/mg	Hu et al., 2023
	Signal peptide engineering, Replacing lid domain from <i>T. lanuginosus</i>	Monoacylglycerol lipase in <i>Thermococcus onnurineus</i>	Specific activity of 400 U/mg	Soni et al., 2019
	Signal peptide engineering	<i>Micrococcus luteus</i> acidic lipase in <i>E. coli</i>	Specific activity of 15.21 U/mg	Nur et al., 2023
		<i>K. marxianus</i> lipase LIP3 in <i>S. cerevisiae</i> BY4742	Secretion of lipase with a specific activity of 1.36 U/mg	Martínez-Corona et al., 2024
	Signal peptide engineering (MSS signal peptide)	<i>K. marxianus</i> lipase YJR107W in <i>S. cerevisiae</i> BY4742	Secretion of lipase with a specific activity of 1.25 U/mg	Martínez-Corona et al., 2024
Transportation	Signal peptide engineering (E7AT signal peptide)	<i>Stenotrophomonas maltophilia</i> lipase in <i>E. coli</i>	Lipase activity of 16.67 U/mL	Gao et al., 2025
	Codon optimization, Signal peptide engineering, co-expression of <i>N</i> -acetyltransferase Mpr1	<i>Proteus vulgaris</i> K80 Lipase in <i>P. pastoris</i>	~2-fold improvement in lipase secretion and activity at 117 U/mL	Yan et al., 2025
	Codon optimization, Promoter optimization (PGCW14), Signal peptide engineering (MFA), Co-expression of <i>Vitreoscilla</i> hemoglobin	<i>T. lanuginosus</i> lipase TLL in <i>Pichia pastoris</i>	29-fold increase in lipase activity	Xu et al., 2023
	Surface Display	<i>C. acetobutylicum</i> NJ4 lipase in <i>E. coli</i>	Lipase displayed on cell surface	Lu et al., 2023b
		<i>P. fluorescens</i> lipase in <i>E. coli</i>	Lipase with >10-fold activity	Baek et al., 2010

be improved by rebuilding the lid using site-directed mutagenesis (Syal et al., 2017; Panizza et al., 2013; and Wang et al., 2021b). By introducing cysteine residues at positions 86 and 255, a disulfide bond was formed, creating a biochemical switch to control the lid conformation of *T. lanuginosus* lipase (Fig. 4B). Upon the addition of the reducing agent tris (2-carboxyethyl) phosphine, the disulfide bond was cleaved, triggering the lid to transition to an open state. This conformational change resulted in a seven-fold increase in the binding rate of the unlocked lipase. This innovative approach provides a foundation for designing tunable lipases with controlled activation, offering significant potential for industrial and biotechnological applications. (Skjold-Jørgensen et al., 2017). Feruloyl esterase A from *A. niger* features a distinctive N-glycosylation site at Asn79, which plays a crucial role in maintaining the lid structure in its open state (Sánchez-González et al., 2012). Manetti et al. (2000) confirmed the important roles of Phe 344 and Phe 345 in

C. rugosa lipase activity. Lid swapping resulted in the improvement of thermostability and enzymatic activity, such as the segments of the Lip4 C-terminus (from residues 374, 396, 417, or 444 to residue 534) were exchanged with the equivalent regions of the Lip1 isoform from *C. rugosa* (Hung et al., 2011).

Additionally, Soni et al. (2019) broadened the substrate specificity of the lipase from *Thermococcus onnurineus* by substituting the first 61 N-terminus amino acid residues of the lid with a 118-residue lid domain derived from *T. lanuginosus*. This modification allowed the enzyme to effectively hydrolyze triglycerides instead of just monoglycerides (Fig. 4C). The likely reason is that the introduced *T. lanuginosus* lipase lid domain enabled interfacial activation, allowing the active site to accommodate larger substrates. Additionally, the replacement also induced conformational changes, including the repositioning of catalytic residues, and further enhanced substrate accessibility and catalytic

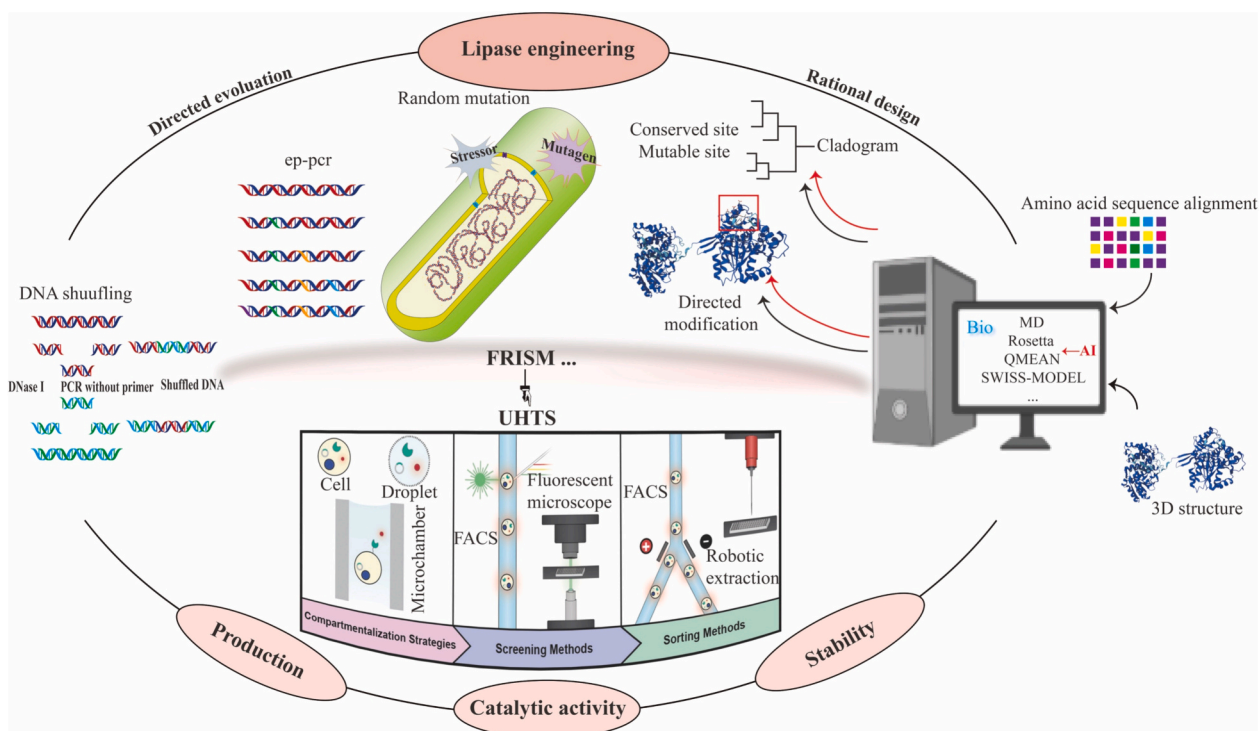


Fig. 3. Lipase engineering through directed evolution, rational design, and their combination. There are three common methods related to directed evolution, DNA shuffling, ep-PCR, and random mutation. For the rational design, sequence-based method and structure-based method are widely employed. The focused rational iterative site-specific mutagenesis (FRISM) is an example of the combination of directed evolution and rational design. AI is used in predicting and identifying functional domains to streamline lipase design and optimization. High-throughput screening (HTS) is used to find desired mutants from the large pool generated in directed evolution. [HTS insert adapted from Longwell et al., 2017 with minor modifications]. FACS, fluorescence-activated cell sorting.

Table 5

Protein engineering techniques employed to enhance lipase properties.

Target	Lipase	Method	Outcome	Reference
Catalytic ability	<i>T. lanuginosus</i>	Domain replacement, lid swapping	Broadened the substrate specificity	Soni et al., 2019
	<i>Y. lipolytica</i>	Site-directed mutation	Broadened the substrate specificity	Syal et al., 2017
	<i>Pseudomonas</i> sp.		Increased activity by 5 to 11-fold	Panizza et al., 2013
	<i>R. chinensis</i>		Increased hydrolytic activity by 7-fold	Wang et al., 2021b
	<i>C. antarctica</i>		Increased enzyme specificity	Xu et al., 2019
	<i>Trichothecium roseum</i>		Increased activity towards substrates with 12, 16, and 18-carbon side chain	Ma et al., 2022
	<i>C. antarctica</i> lipase A		Increased activity towards tertiary alcohol by 1.5-fold	Wagner et al., 2024
	<i>C. antarctica</i> lipase A	ep-PCR, random mutagenesis	14 times higher ability to hydrolyze triacylglycerol	Glód, 2017
	<i>T. lanuginosus</i>	Introducing disulfide bond, lock the open conformation of lids	Seven-fold increase in substrate binding rate	Skjold-Jørgensen et al., 2017
	<i>Geobacillus thermocatenulatus</i>	Site-directed oriented immobilization	High enantiomeric excess (ee > 99 %)	Godoy et al., 2013
Thermostability	<i>B. subtilis</i>	Loop scanning and site-saturation mutagenesis	Increased melting temperature by 1–6 °C	Ahmad et al., 2012
	<i>R. chinensis</i> lipase in <i>P. pastoris</i>	Directed evolution, ep-PCR and DNA shuffling	22 °C rise in melting temperature	Yu et al., 2012
	<i>C. rugosa</i> lipase	Site-directed mutation	7.4–9.3 °C rise in melting temperature	Li et al., 2018
	<i>Malassezia globosa</i>	Site-directed mutation	14 °C rise in melting temperature, and 1154.3-fold in the half-life	Li et al., 2022
	<i>B. subtilis</i> lipase A in <i>E. coli</i>		8-fold catalytic turnover in DMSO	Yedavalli and Rao, 2013
Solvent stability	<i>T. lanuginosus</i>	Site saturation mutagenesis	6.7-fold longer half-life in methanol	Okamura et al., 2022
	<i>Trichothecium roseum</i>		13-fold higher methanol tolerance	Ma et al., 2023
	<i>B. subtilis</i>		Increased stability in DMSO	Cui et al., 2022
	<i>B. subtilis</i>		Increased stability in ionic liquid	Pramanik et al., 2022
Thermal and solvent stability	<i>G. stearothermophilus</i>	Site-directed mutation	Enhanced thermostability by 87-fold in buffer and in 50 % methanol	Dror et al., 2015
Thermal, solvent, and pH stability	<i>G. zalihae</i> T1	Site-directed mutation	Increased stability	Ishak et al., 2019
Thermal and pH stability	<i>P. plecoglossicida</i> S7	Site-directed mutation	Lipase activity increased by 1.2-fold	Choudhary et al., 2023

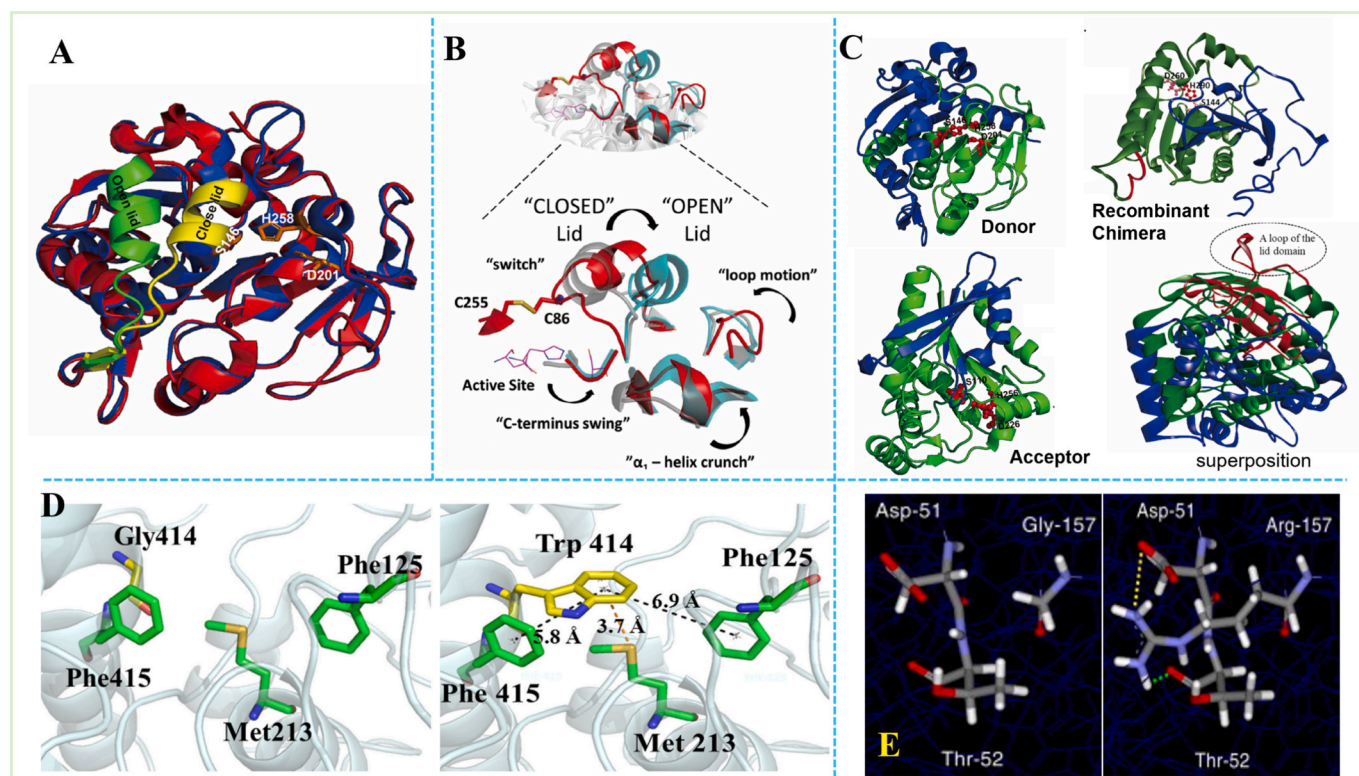


Fig. 4. Conformation of lipase with lid and site mutagenesis to improve its catalytic activity. A. Conformational changes of *T. lanuginosus* lipase with lid in open (green, 1EIN) and closed (yellow, 1DT3) states [obtained from the PDB database and visualized using PyMOL]. B. Conformational changes of *T. lanuginosus* lipase lid in locked (gray), closed (red), and open (pale blue) states after introducing a disulfide bond. [Adapted from Skjold-Jørgensen et al., 2017]. C. Lid swapping between lipase from *Thermococcus onnurineus* (acceptor) and *T. lanuginosus* (donor) to create the recombinant chimera, with the superposition of the chimera and acceptor showing conformational changes (bottom right). The acceptor (6EIC) and donor (1DT5) are obtained from the PDB database and visualized using PyMOL. All catalytic residues are labeled with their single-letter code and corresponding position. [Adapted from Soni et al., 2019]. D. G414W mutation in *C. rugosa* lipase Lip1. The mutant (right) shows aromatic cluster between mutated amino acid Trp414 and three neighboring amino acids Phe415, Met213, and Phe125 [Adapted from Zhang et al., 2016a]. E. G157R mutation in *P. aeruginosa* LST-03 lipase causing the formation of salt bridge (yellow) and hydrogen bond (green) between Arg-157 and Asp-51 and Thr-52, respectively, in the mutant lipase (right). [Adapted from Kawata and Ogino, 2010]. (For interpretation of the colour in this figure legend, the reader is referred to the web version of this article.)

efficiency. Wagner et al. (2024) introduced mutations in amino acid residues surrounding the active site to enhance the lipase's acceptance of bulky tertiary alcohols, resulting in a 1.5-fold increase in catalytic activity. For well-known lipases, modulating enzyme activity commonly involves altering substrate accessibility, which can be achieved not only by modifying residues around the active site but also by engineering the substrate access tunnel (Chong et al., 2023). The glycosidic chains also play a vital role in lipase enzyme activity. Brocca et al. (2000) discovered that both Asn314Gln and Asn351Gln could affect the activity of *C. rugosa* Lip1 by changing the interaction between the sugar chain and residues of the lid. In addition, mutating the oxyanion hole residue Phe148 (close to active center) to Leu in lipase 12 from *Y. lipolytica* resulted in a 2-fold increase in the hydrolytic activity towards olive oil (Kumari and Gupta, 2013). Collectively, beyond the active site, the lid conformation is crucial for catalytic activity—not only the lid itself but also other structural components that can indirectly influence its configuration.

4.2. Enhanced lipase stability

Poor thermal stability can restrict a lipase's industrial applications. Improving a lipase's thermal stability usually also improves its catalytic performance. Santarossa et al. (2005) used rational design to improve the thermostability of *Pseudomonas fragi* lipase. They discovered that three polar residues (T137, T138, and S141) in the lid region of the cold-adapted lipase enzyme significantly affected the lipase's

thermostability. Introducing T137V and T138N mutations that replaced threonine with conserved amino acids found in thermostable homologous lipases resulted in prolonged retention of high enzymatic activity. In contrast, the S141G substitution markedly decreased thermostability, likely due to the increased flexibility conferred by the glycine residue. Besides the lid domain, the substitution of the oxyanion hole residue Gln114 with Leu in *Geobacillus zalihae* lipase led to a 10 °C elevation in the enzyme's optimal temperature, which might be attributed to increased hydrophobic interactions near the active site (Wahab et al., 2012).

When knowledge about the enzyme is limited, directed evolution is often used to develop desirable mutants. Yu et al. (2012) employed two consecutive rounds of error-prone PCR (ep-PCR) and DNA shuffling to increase *Rhizopus chinensis* lipase thermostability. This method improved the protein surface hydrophilicity and polarity, which led to a 22 °C rise in its melting temperature and a 46-fold increase in the enzyme half-life at 60 °C. They also discovered that adding a disulfide link to the hinge area of the lid boosted thermostability and extended the half-life by 11-fold at 60 °C. Rational design helps narrow down target sites, offering significant advantages in saving time and reducing laboratory workload. The acyl-binding residue Gly414 was identified, through multiple sequence alignments and structural analyses, as a target to increase the thermostability of *C. rugosa* lipase Lip1 (Zhang et al., 2016a). Four mutants (G414A, G414M, G414H, and G414W) were identified through saturation mutagenesis, which involves mutating a specific position to all other natural amino acids; among them, G414W

exhibited the most significant improvement with a 6.2-fold increase in half-life at 60 °C. Through modeling and docking analysis, this enhanced thermostability was attributed to strengthened interactions, including the formation of hydrophobic clusters (like aromatic cluster) and hydrogen bonds within the acyl-binding tunnel (Fig. 4D). In addition, F344I, F434Y, F133Y, F121Y, and D457F were also identified to enhance Lip1 stability (Zhang et al., 2016b; Li et al., 2018). The strategy of active center stabilization was employed to investigate the impact of mutating residues surrounding the active site on protein stability (Zhang et al., 2016b). Based on B-factor rankings, flexible residues were identified and selected for site-saturation mutagenesis. Through three-tier high-throughput screening, several beneficial mutations (F344I, F434Y, F133Y, and F121Y) were discovered, which resulted in a 40-fold increase in the enzyme half-life at 60 °C and a 12.7 °C increase in its melting temperature (T_m). Although mutations near the active center often present a trade-off between thermostability and catalytic activity, in this case, no loss in catalytic efficiency was observed. The results suggest that targeting flexible residues for mutation is a promising strategy to enhance stability without compromising enzyme activity.

In addition, Li et al. (2018) employed Rosetta ddg_monomer, FoldX, and I-Mutant 3.0 to computationally predict the effects of mutations on thermostability. By integrating these three methods, it became easier to screen and identify potential mutation candidates for stability enhancement. Collectively, both the lid and the domain surrounding the active center are hotspots for modulating lipase thermostability. However, mutations in residues near the active center should be approached with caution due to the potential trade-off between thermostability and catalytic activity.

Most lipases show interfacial activation, so the stability of lipases in organic solvents is particularly important. Kawata and Ogino, 2009 used ep-PCR for site-directed mutagenesis (S155L, G157R, S164K, S194R, and D209N) to increase the half-life (twice longer) of *Pseudomonas aeruginosa* lipase in DMSO. Through 3D analysis, these changes improved the packing of the lipase's hydrophobic core by introducing additional salt bridges and hydrogen bonds, which were associated with enhanced stability in organic solvents (Fig. 4E) (Kawata and Ogino, 2010). Hydrogen bonds and ionic interactions were also found to be crucial in maintaining the *Geobacillus zalihae* lipase stability under challenging conditions, and site-directed mutations (D43E, T118N, E226D, E250L, and N304E) in these regions enhanced both the lipase melting temperature and its stability in various organic solvents, as well as under acidic and alkaline pH conditions (Ishak et al., 2019). Similarly, Okamura et al. (2022) introduced hydrogen bond and salt bridge by replacing hydrophobic with hydrophilic amino acid, resulting in 6.7-fold longer half-life in 50 % methanol. Using site saturation mutagenesis to examine 18,000 clones of all 91 amino acids in the loop region, Yedavalli and Rao (2013) discovered that a mutant with six mutations (I12L, W42L, A68S, P119S, L140F, Y139K) had an eight-fold greater catalytic activity in 60 % DMSO. They suggested that lipases can be made more stable in organic solvents by changing their surface residues, particularly those in the loop region. Dror et al. (2015) reported that substituting charged surface residues with hydrophobic ones increased the stability of *Geobacillus stearothermophilus* T6 lipase by 87-fold in buffer and 70 % in methanol. Through surface charge engineering, *B. subtilis* lipase stability in ionic liquid and DMSO was also increased significantly (Pramanik et al., 2022; Cui et al., 2022). It is clear that surface residues play a crucial role in maintaining lipase stability in organic solvents, possibly due to the enzyme's inherent property of interfacial activation. Enhancing hydrophobicity, alongside the introduction of salt bridges and/or hydrogen bonds, represents an effective strategy for improving lipase stability.

4.3. Directed evolution for protein engineering

The rational design approach for enzyme engineering typically relies on a deep understanding of enzyme structure and the intricate

connections between sequence, structure, and function or mechanism. There are alternative strategies for enzyme optimization and engineering that do not necessarily require such detailed prior knowledge. These approaches are often referred to as “directed evolution” methods, which are particularly useful when the enzyme's structure-function relationship is not well understood or when one aims to generate novel enzymes with improved properties. Error-prone PCR (ep-PCR) and DNA shuffling are among the most widely used techniques for generating variant libraries in directed evolution studies. Glód (2017) utilized ep-PCR to create *C. antarctica* lipase A mutants, enhancing their fatty acid selectivity and boosting their lipolytic activities. A mutant with a 14 times higher activity to hydrolyze triacylglycerols was obtained.

Nevertheless, directed evolution generates a plethora of mutants, necessitating the utilization of high-throughput screening (HTS) methods. Several emerging protein engineering platforms with differing compartmentalization strategies, screening methods, and sorting techniques have been developed to screen enzyme libraries (Fig. 3). Cell-based compartmentalization, droplet-based systems, and microchamber-based platforms are the three most popular compartmentalization strategies. Both fluorescence-activated cell sorting (FACS) and fluorescent microscopy methods ensure accurate screening. Moreover, FACS can also be used for mutant sorting, offering an efficient approach to isolate individual mutants. By combining these methods in different steps, numerous screening strategies can be formulated, achieving screening rates that exceed 10⁵ variants per hour. (Longwell et al., 2017). Consequently, given the ongoing advancements in high-throughput screening techniques, directed evolution is expected to remain the preferred choice for developing lipases tailored for commercial applications.

In addition to the advancement of high throughput screening methods, reducing mutant libraries by concentrating on key amino acids, such as those in the active site, shows great potential. Xu et al. (2019) employed a strategy known as “focused rational iterative site-specific mutagenesis” (FRISM), targeting residues within the binding pocket of *C. antarctica* lipase B. This approach involved iterative site-specific mutagenesis using a carefully selected subset of rationally chosen amino acids, enabling the creation of ultrasmall mutant libraries that required minimal screening to achieve high stereoselectivity. By integrating directed evolution with rational design, researchers can efficiently and precisely reduce the scale of mutant libraries, making it a promising approach for protein engineering.

4.4. In silico analysis aided protein engineering

In silico analysis using molecular dynamics (MD) simulation is a promising tool to explore the biocatalytic process and help the design of improved lipase mutants. Bhattacharjee et al. (2023) employed MD and dimensionality reduction to reveal the effects of immobilization on the lid state as well as the relationship between the lid state and lipase activity. Chen et al. (2020) also employed MD to reveal the relationship between temperature and the lipase lid, which is closely related to the catalytic activity. The lid plays a vital role in the regulation of lipase activity and stability, and MD simulations, guided by a combination of bioinformatics and processing conditions, can be regarded as a platform for redesigning the lid and enabling the creation of targeted lipases with novel desired properties.

Ancestral sequence reconstruction (ASR) is another simple and efficient method to optimize the protein properties. Using an alignment of homologous proteins, ASR methods estimate a phylogenetic tree and the sequences of ancestral sequences in that tree. Proteins generated by ASR tend to be very thermostable and have broadened substrate scope (Zhao et al., 2025). Based on the spatial structure of *Thermobaculum roseum* lipase, Ma et al. (2022) employed ASR to reduce steric hindrance to increase the catalytic activity on longer chains of *p*-nitrophenyl esters. Additional structure-guided computational tools, including homology modeling (SWISS-MODEL, YASARA, and MODELLER), model validation

(PROCHECK, VERIFY 3D, QMEAN, ERRAT2, and MOLPROBITY), molecular docking (AUTODOCK VINA, and AUTODOCK), and MD simulation (GROMOS54, HARMM36, and AMBER14) commonly used in lipase engineering can be found in a recent review paper (Vardar-Yel et al., 2024). In practice, combining the above models with rational design is more effective. For example, the AutoDock Vina program was used to evaluate site-saturation mutagenesis results and assist in designing fatty acid selectivity (Moharana and Rao, 2020).

The Framework for Rapid Enzyme Stabilization by Computational Libraries (FRESCO), a combination of MD simulation and rational design, is another widely used system for enzyme stabilization. Basically, stabilizing mutations are predicted using tools like Rosetta or FoldX, and filtered by MD simulations to remove dynamically unstable variants. After excluding functionally disruptive sites, selected mutants are tested for thermostability. Li et al. (2022) used FRESCO to design single point mutations folding free energy $\Delta\Delta G^{\text{Fold}}$ and then narrow down the libraries by visual inspection and MD simulation with Yet Another Scientific Artificial Reality Application (YASARA). They also introduced disulfide bonds to further increase thermostability. The resulting *Malassezia globosa* lipase (SMG1) showed significant improvement on melting temperature, optimal temperature, and half-life. Tan et al. (2022) employed the B-FITTER program (Watanabe et al., 2021) to compute B factors, which measure the flexibility of each residue in a structure. Then, a total of 8 amino acids were selected to conduct saturation mutagenesis, and the resulting *C. antarctica* lipase B V306N displayed 31 % higher activity than wild type with 15 % (v/v) addition of methanol.

4.5. Machine learning assisted protein engineering

ML offers powerful capabilities in protein engineering. Ge et al. (2023) reviewed AI and ML tools for lipase production and engineering. They compared various modeling algorithms, including commonly used wavelet neural networks (WNN) and convolutional neural networks (CNN), highlighting their respective advantages and limitations. The CNN model is the most widely investigated among different deep neural networks. With the input of a 3D cube of dimensions $20 \times 20 \times 20 \text{ \AA}^3$ centered on the lipase catalytic residues, Galmés et al. (2021) trained a CNN model to identify conserved structural patterns associated with amidase activity. Their study showed that the CNN can help to quickly locate the functional domain of an enzyme. Fard Masoumi et al. (2014) applied and compared ANNs and WNNs in studying lipase-catalyzed synthesis of triethanolamine. A total of 5 variables including enzyme amount, reaction, temperature, substrate molar ratio, and agitation speed were used to train both models to predict the enzymatic reaction. Their results showed that the WNN model gave a better alignment between predicted values and experimental results. The relatively small dataset (30 experimental points) in the study was probably insufficient for effective CNN training. On the other hand, WNN performed well with limited data due to the compact and localized nature of wavelet basis functions. This study highlighted the importance of selecting an appropriate model tailored to the specific case. Also, high prediction accuracy often requires substantial computational power, which can limit AI's broader application.

Ma et al. (2023) employed ML-assisted directed evolution to increase the tolerance of *Proteus mirabilis* lipase to methanol. They first performed multi-site saturation mutagenesis to generate a diverse set of variants, which were used to construct a training library for machine learning. The multiple sequence alignment (MSA) Transformer (Rao et al., 2021) was employed to extract evolutionary and positional information from the protein sequences, which were then used as input features for model training. A total of 22 regression models, implemented using Python packages such as Keras, Scikit-learn, and XGBoost, were constructed and trained. Based on tenfold cross-validation performance, the top three models (XGB-Tree-Tweedie, XGB-Tree, and sklearn-regressor-BaggingRegression) were selected for further evaluation. Finally, the

mutant G202N/K208G/G266S consistently scoring highest across predictions from the leading models was identified. This mutant exhibited a 13-fold increase in methanol tolerance compared to the wild type. MD simulations revealed that the reconstruction of hydrogen bonds within the structure of the engineered protein contributed to the enhanced enzyme stability.

Recently, Zheng et al. (2025) introduced a novel enzyme engineering framework called iCASE (isothermal compressibility-assisted dynamic squeezing index perturbation engineering), which integrates molecular dynamics, computational feature extraction, and ML to simultaneously improve the thermostability and activity of 10 industrial enzymes including *Rhizomucor miehei* lipase. A supervised structure-based random forest regression model was developed to predict enzyme function and fitness, using five key input features derived from molecular simulations: isothermal compressibility (βT), dynamic squeezing index (DSI), root mean square deviation (RMSD), radius of gyration (R_g), and solvent accessible surface area (SASA). This iCASE-based strategy overcomes the traditional stability-activity trade-off and can accelerate the fitness evolution of industrial enzymes.

While ML can be effective with only a few dozen to a few hundred data points, obtaining such datasets is still challenging, especially for specialized applications like lipase-catalyzed esterification. To date, most studies have focused on improving lipase hydrolysis activity, thermostability, or solvent tolerance, whereas efforts to engineer lipases for enhanced short-chain ester synthesis remain limited. This presents a significant challenge for laboratory research and underscores the urgent need for HTS technologies. Furthermore, ML requires high-quality data to train effective models. HTS technologies can facilitate the collection of training data sets needed to develop a valid ML model. The integration of HTS with ML thus holds great promise for advancing protein engineering.

5. Lipase immobilization

Enzyme immobilization enhances its recyclability, stability, and activity (Liu et al., 2018) and thus can reduce lipase cost in industrial applications. Various methods have been used for lipase immobilization, including physical (adsorption, encapsulation, and entrapment), chemical (covalent binding and cross-linking), and ionic (ionic exchange). These immobilization methods

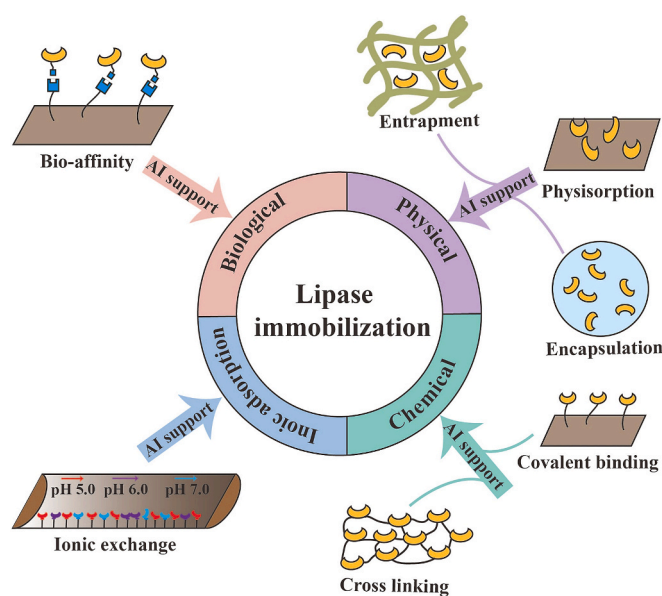


Fig. 5. Schematic illustration of commonly used methods for lipase immobilization. AI is often used to aid carrier design and optimization of the immobilized lipase performance.

are illustrated in Fig. 5. Table 6 lists notable lipase immobilization studies reported in the literature. In this section, we provide a brief overview of methods used for lipase immobilization, followed with detailed discussion on novel enzyme supports for enhancing lipase loading, activity, and stability in various industrial applications.

5.1. Enzyme immobilization methods

Both physical and chemical methods are widely used to immobilize lipase on different supports (Fig. 6A). Ionic adsorption immobilizes enzymes through electrostatic interactions with oppositely charged ion exchange resins. Bhushan et al. (2008) and Badgujar and Bhanage (2016) applied the entrapment method to immobilize lipase on different supports, including calcium alginate beads and hydroxypropyl methyl cellulose. However, lipase entrapment usually had a poor enzyme loading, which might be limited by the available active site restricted by the embedding medium. Zhao et al. (2017) entrapped *Candida* sp. 99–125 lipase in calcium alginate beads and achieved a high enzyme loading density that led to spatial overlap of the lipase molecules (Fig. 6B). Tran et al. (2012) and Silva et al. (2014) immobilized lipase by physisorption on particles, achieving better performance than the entrapment method. Chen et al. (2022) and Shomal et al. (2022) used encapsulation to immobilize lipase in polymers, which gave an enzyme loading of 26.9–43.1 mg per g of the polymers. Combining encapsulation in calcium alginate gel with adsorption on ionic liquid-modified silica carriers gave a high enzyme loading of 186 mg lipase per gram of support with good thermal stability (Zou et al., 2018), whereas only 4.5 mg lipase per gram of support was achieved via ionic adsorption on 2-methoxyethylamine agarose (Mendes et al., 2011).

Lipase immobilization by covalent binding on an epoxy-SiO₂-PVA support (Mendes et al., 2011) and PVDF membrane (Kuo et al., 2014) gave a low enzyme loading of 3 mg and 1.71 mg/g support, respectively. Meanwhile, covalent binding resulted in 676.2 U of lipase immobilized on PVA-co-PE nanofibrous membranes (Zhu and Sun, 2012) and 60.2 U of lipase on magnetic Fe₃O₄ chitosan nanoparticles (Wang et al., 2015). In general, covalent binding gave a low enzyme loading and was not the

best way for lipase immobilization.

A precipitation-cross-linking method was developed to immobilize lipase on magnetic cellulose nanocrystals at a high enzyme loading of 82.2 mg/g with 95.9 % activity recovery (Cao et al., 2016). A much higher enzyme loading of 250 mg/g support with ~95 % efficiency was developed for lactase by using poly-anionic polyethyleneimine (PEI) to form PEI-enzyme aggregates adsorbed on cotton fibers (Albayrak and Yang, 2002). The high enzyme loading led to stable reactor performance at a 400-fold higher reactor productivity with excellent process stability. Similarly, *Candida* sp. 99–125 lipase was adsorbed onto textile membrane coated with polyethylene glycol-6000, glutin, lecithin, and MgCl₂ for biodiesel production (Lu et al., 2007; Tan et al., 2010).

Finally, bio-affinity immobilization, which is more efficient than chemical and physical methods, has also been developed for lipase. Bandikari et al. (2018) overexpressed lipase fused with the cellulose binding domain (CBD), which showed 37 % higher enzyme loading capacity than the enzyme without CBD. Similarly, Abdelhamid et al. (2022) fused *B. subtilis* lipase A with two silaffin peptides Sil3K (Fig. 6C), which resulted in a 70-fold increase in its catalytic activity compared to the wild type lipase. Moreover, the immobilized lipase was evenly distributed across the support surface. By introducing fusion peptides to the lipase, its orientation on the support surface becomes more controlled, thereby reducing the impact of steric hindrance. The immobilized lipase retained 98 % of its catalytic activity after 10 consecutive cycles of use. Atiroglu et al. (2023) utilized the bio-affinity of the biomacromolecule lactoferrin as a support to create a favorable microenvironment for lipase immobilization. The lactoferrin-lipase complex was subsequently immobilized onto a metal-organic framework (MOF) through coordination between lactoferrin and the metal ions within the MOF. This significantly improved the lipase activity and thermal stability. Therefore, bio-affinity immobilization can be achieved not only by fusing lipase with specific binding motifs, but also by incorporating bioactive macromolecules.

Table 6
Methods and supports used for lipase immobilization.

Method	Lipase Source	Support	Enzyme loading (per g support)	References
Entrapment	<i>Arthrobacter</i> sp.	Calcium alginate beads	5 U	Bhushan et al., 2008
	<i>P. fluorescence</i>	Hydroxypropyl methyl cellulose	N/A	Badgujar and Bhanage, 2016
	<i>C. viswanathii</i>	octyl Sepharose	109.2 U	Almeida et al., 2018
	<i>Burkholderia</i> sp.	Methyl-grafted Fe ₃ O ₄ -SiO ₂ nanocomposites	29.5 mg lipase	Tran et al., 2012
Physisorption	Porcine pancreatic	Poly-hydroxybutyrate particles	292.8 U	Silva et al., 2014
	<i>R. miehei</i>	Cu(NO ₃) ₂ ·3H ₂ O, benzene-1,3,5-tricarboxylic acid (CuBTC)	N/A	Liu et al., 2024
	<i>T. lanuginosus</i> & <i>C. antarctica</i>	Mesoporous MIL-88A	116.4 mg/76.7 mg	Li et al., 2024a
	<i>A. oryzae</i>	ZIF and polydimethylsiloxane	N/A	Zhong et al., 2024
Encapsulation	<i>T. lanuginosus</i>	Nucleotide-hybrid metal coordination polymers	43.1 mg lipase	Chen et al., 2022
	<i>A. oryzae</i>	Zeolitic imidazolate frameworks-67	26.9 mg lipase	Shomal et al., 2022
Ionic adsorption and Encapsulation	Porcine pancreatic	Ionic liquid-modified SiO ₂ and calcium alginate	186 mg lipase, 87 U	Zou et al., 2018
Ionic adsorption	<i>P. camembertii</i>	2-Methoxyethylamine agarose	4.5 mg lipase	Mendes et al., 2011
	<i>P. camembertii</i>	Epoxy-SiO ₂ -PVA	3 mg lipase	Mendes et al., 2011
	<i>C. rugosa</i>	Poly (vinyl alcohol-co-ethylene) (PVA-co-PE) nanofibrous membranes	676.2 U	Zhu and Sun, 2012
	<i>C. rugosa</i>	Polyvinylidene fluoride (PVDF) membrane	1.7 mg lipase	Kuo et al., 2014
	<i>T. lanuginosus</i>	Magnetic Fe ₃ O ₄ chitosan nanoparticles	60.2 U	Wang et al., 2015
	<i>C. antarctica</i>	Multi-walled carbon nanotubes, and pickering emulsion	30.1 mg lipase	Wang et al., 2018
Covalent binding	<i>R. oryzae</i> and <i>C. rugosa</i>	Multi-walled carbon nanotubes, and four-arm polyethylene glycol amine polymer	996.26 U *	Abdulmalek et al., 2021
	<i>C. rugosa</i>	Magnetic multi-walled carbon nanotubes, and amine functionalized polyamidoamine	8672.61 U	Kaur et al., 2024a
Cross-linking	<i>P. cepacia</i>	Magnetic cellulose nanocrystals	82.2 mg lipase	Cao et al., 2016
	<i>G. stearotheophilus</i>	Magnetic cellulose nanosphere	560 mg lipase	Bandikari et al., 2018
Bioaffinity	<i>C. antarctica</i>	Lactoferrin (LF) hybrid graphene oxide (GO), Pt Nanoparticles and Metal-organic frameworks (MOF-74)	N/A	Atiroglu et al., 2023

Note: Data marked with * in the table were calculated by us based on the results from the original research.

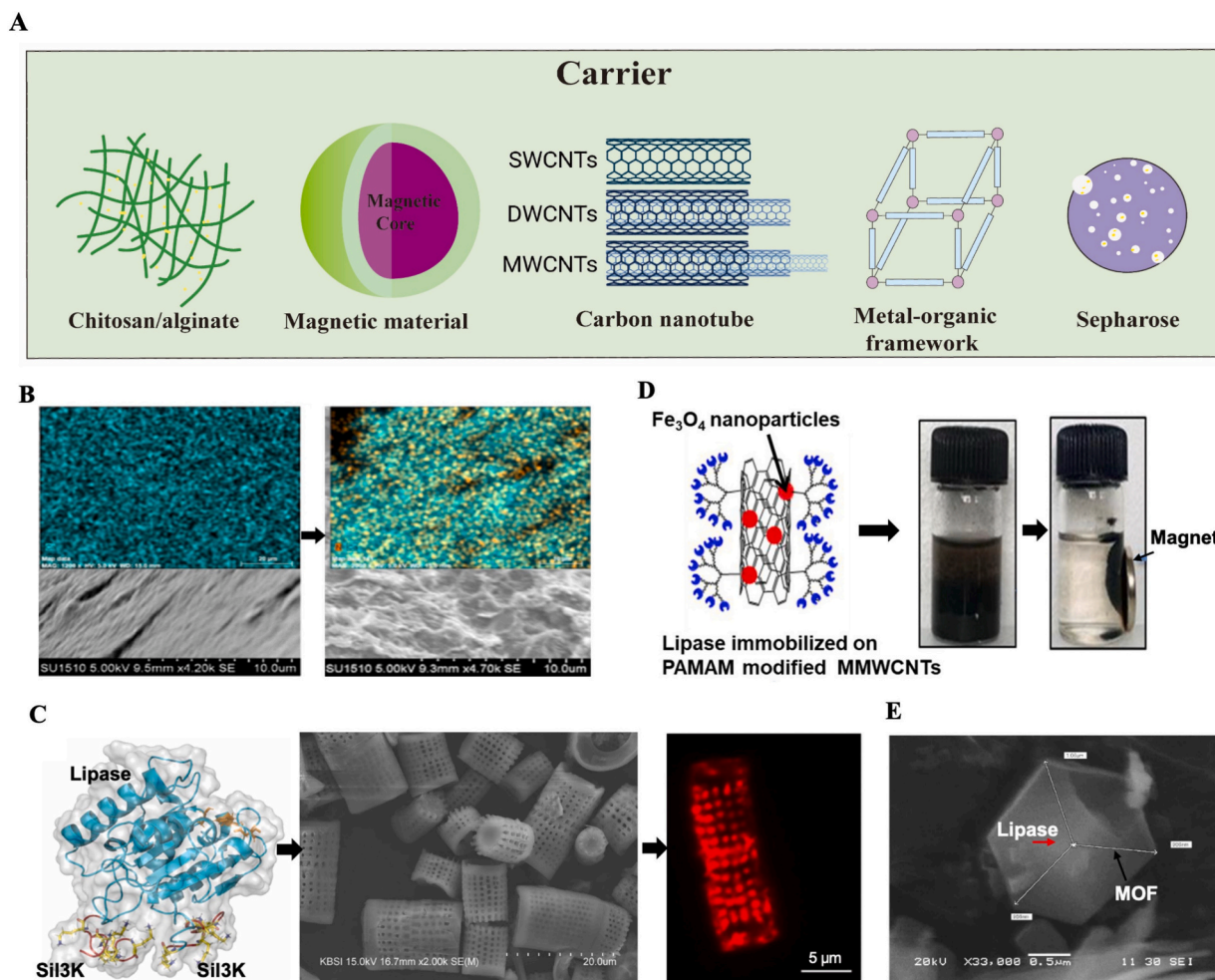


Fig. 6. Carriers used in lipase immobilization. (A) Schematic illustration of different carriers used in lipase immobilization are shown. [SWCNTs, single-walled carbon nanotubes; DWCNTs, Double-walled carbon nanotubes; MWCNTs, multi-walled carbon nanotubes]. (B) Microscopy images of Ca-alginate gel (Left) with entrapped *Candida* sp. 99-125 lipase (Right) [Source: Zhao et al., 2017]. (C) *B. subtilis* lipase fused with diatom-derived silaffin tag Sil3K (Left) to enable immobilization on biosilica (Middle), visualized after staining with Cy5-NHS (Right) [Source: Abdelhamid et al., 2022]. (D) *C. rugosa* lipase immobilized on magnetic MWCNTs [Source: Kaur et al., 2024a]. (E) Scanning electronic microscopy image of MOF (ZIF-8) with *A. niger* lipase [Source: Nadar and Rathod, 2018].

5.2. Enzyme support materials

The enzyme support material is crucial for lipase loading and catalytic performance. Chitosan stands out as one of the prevalent matrices among various supports. Various advancements have been made to enhance immobilization techniques, including refining immobilization methods, integrating them with complementary approaches, chemically and physically modifying chitosan, employing co-immobilization strategies, and utilizing protein engineering (Rafiee and Rezaee, 2021).

Nanomaterials offer notable advantages, including a larger surface area, improved dispersibility, and reduced mass transfer limitations. First, carbon nanotube is a promising carrier for lipase due to its stability and biocompatibility (Zdarta et al., 2018). Extensive research efforts have been done to increase active sites by carbonylation (Deep et al., 2015) and dendrimer modification (Fan et al., 2016; Fan et al., 2017). Kaur et al. (2024a) employed magnetic multi-walled carbon nanotubes (MWCNTs) to immobilize *C. rugosa* lipase (Fig. 6D). The immobilized enzyme exhibited high enzymatic activity, exceeding 8000 U/g of support, and good stability, retaining 65.2 % of its activity even after 11 cycles of reuse. Abdulmalek et al. (2021) co-immobilized the lipases from *R. oryzae* and *C. rugosa* with the carbon nanotube modified with a four-arm polyethylene glycol amine polymer, which showed 34.5-fold activity and significantly improved thermal stability. Meanwhile, they also employed the magnetic nanoparticles Fe_2O_4 to recover the lipase,

which displayed 78.55 % biodiesel yield after ten reuse cycles. In addition, carbon nanotubes were used to stabilize a Pickering emulsion system encapsulating lipase that achieved ~95 % conversion rate of biodiesel with up to 10 times reuse without significant activity loss (Wang et al., 2018). However, carbon nanotubes have several disadvantages, including high cost, lower recovery rate, and limited active sites on the surface of carbon nanotubes. Attaching magnetic nanoparticles such as capped iron oxide, Co, and CoPt to carbon nanotubes was shown to be a feasible way to recover immobilized enzymes for reuse that can potentially greatly reduce the enzyme cost (Georgakilas et al., 2005).

Metal-organic frameworks (MOFs), which exhibit controllable morphology and strong affinity for enzymes, are also widely used for lipase immobilization (Kuang et al., 2023; Zhong et al., 2021). Patil et al. (2023) offered an in-depth review of MOF-mediated lipase immobilization. Zeolitic imidazolate frameworks (ZIFs), a subclass of MOFs, are among the most studied carriers for lipase immobilization. Among the ZIFs materials, ZIF-8 and ZIF-L are two most popular ones because of their negligible cytotoxicity (Shomal et al., 2021). Nadar and Rathod (2018) encapsulated *A. niger* lipase in ZIF-8 (Fig. 6E), which showed a 3.2-fold increase in its half-life compared to the free lipase. However, its residual activity was only 54 % after 5 consecutive cycles of use. In addition, Liu et al. (2024) synthesized a series of CuBTC derived dual-ligand MOFs and used them for the immobilization of *Rhizomucor*

miehei lipase with a 66.9 % increase in the specific activity. A mesoporous MIL-88 A with a pore size larger than 10 nm was also developed for enzyme immobilization (Li et al., 2024a). MIL-88 A is a low-cost, highly versatile material with good flexibility, biocompatibility, water stability, and magnetic properties. When used as the carrier for *C. antarctica* lipase B, MIL-88 A increased the enzyme loading, specific activity, and reusability. In general, coordination competition and spatial confinement are two common drawbacks in using MOFs as enzyme supports. However, strategies such as incorporating bioactive molecules like lactoferrin can help mitigate these drawbacks and further enhance the applicability of MOFs in lipase immobilization.

Molecular simulations used to investigate how lipases with different lid structures interact with surfaces of varying hydrophobicity revealed that an optimal hydrophilic/hydrophobic balance facilitated interfacial activation (Zhong et al., 2024). Based on these insights, a novel super-hydrophobic ZIF nanomaterial (ZIFN) was developed, which showed that the modification with hydrophilic tannic acid significantly enhanced the immobilized lipase activity and reusability (Zhong et al., 2024). This study demonstrated the potential of rational surface design in improving enzyme immobilization efficiency.

5.3. Machine learning for improving enzyme immobilization design

It should be noted that the optimal design and selection of support materials for lipase immobilization will depend on the immobilization methods. The traditional immobilization methods have been well developed with limited variations in the optimization of carrier characterization and the parameters involved in enzyme immobilization. AI and ML are thus getting increasing attention for application in improving lipase immobilization. Fatiha et al. (2013) applied ANNs and response surface methodology (RSM) to optimize the immobilization of *C. rugosa* lipase on Amberjet® 4200-Cl. The ratio of enzyme to support, temperature, and buffer concentration were used as input variables and enantioselectivity (E value) as the response index. ANN gave superior performance in capturing nonlinear relationships and complex data patterns, resulting in higher predictive accuracy compared to RSM. Chai et al. (2021) utilized two ML algorithms, namely random forest (RF) and Gaussian process regression, to assess the effects of twelve input variables encompassing both support properties and enzyme characteristics on enzyme immobilization. While the RF model demonstrated superior accuracy in predicting enzyme loading, yield, and reusability, neither model reliably predicted enzyme activity retention. A random forest (RF) model was also employed to optimize the immobilization of glucose oxidase and horseradish peroxidase on the ZIF support, resulting in at least a 1.3-fold improvement in immobilization performance (Liang et al., 2024).

Roura Padrosa et al. (2021) developed CapiPy, a Python-based graphical user interface, to assist in protein immobilization using rule-based bioinformatics approaches rather than traditional ML. The tool accepts protein sequences or structures as input, performs homology modeling and surface residue analysis, and identifies suitable immobilization sites while avoiding catalytic regions. However, CapiPy could become significantly more powerful by integrating with ML. By training predictive models on curated datasets of successful and unsuccessful immobilization cases, ML could help identify optimal immobilization sites, estimate post-immobilization activity, and recommend appropriate carriers or strategies. This would elevate CapiPy from a rule-based assistant to an intelligent, data-driven decision-support platform for enzyme immobilization. Meng et al. (2020) employed the support vector machine (SVM) algorithm to predict polystyrene-binding peptides by inputting sequence-based and physicochemical features derived from peptide sequences, which can be adjusted according to the specific support material.

Finally, it should be noted that protein engineering and enzyme immobilization should be considered together in developing lipases for industrial applications. The current approach employing a sequential

combination of protein engineering and enzyme immobilization is not ideal since enzymes enhanced through protein engineering are not always tailored for optimal functionality as immobilized biocatalysts, making their effectiveness after immobilization unpredictable. Engineering the lipase to allow for controlling enzyme orientation during the immobilization process on a carrier material would be crucial for not only immobilization but also in maximizing the immobilized enzyme activity (Liu et al., 2015). Wang et al. (2016) improved both the immobilization specificity and lipase catalytic performance with an engineered lipase with non-natural amino acids. Optimizing the interactions between the lipase and its support by modifying surface amino acids and/or tailoring the carrier surface to improve affinity thus holds great promise for enhancing the performance of immobilized enzymes. Integrating the immobilization step as an evolutionary pressure in protein engineering can also be a critical point in identifying enzymes in a mutant library specifically tailored for immobilization.

6. Conclusions and future perspectives

Lipases have emerged as highly promising green catalysts with applications across various industries. Their versatility stems from their ability to perform various catalytic functions. Considerable progress has been made in enhancing lipase production, activity, stability, and catalytic specificity through protein engineering and immobilization. However, as industrial demands continue to grow, there is a need for more specific and efficient lipases. Apart from well-known commercial lipases like those from *C. antarctica* and *C. rugosa*, extremophilic lipases from thermophilic, psychrophilic, halophilic, alkaliphilic, acidophilic, and organic solvent-tolerant organisms have been discovered. These extremophilic lipases can serve as templates for modifying common lipases to enhance their performance. Integration of protein engineering and immobilization, where both techniques are selected as inherent evolutionary pressures, holds enormous potential for generating improved biocatalysts for industrial applications. By combining these approaches, lipases can be tailored to meet specific requirements and exhibit enhanced properties. Process optimization is also vital for improving efficiency at various scales of operation. Statistical designs, such as the response surface method and regression analysis, have also been proven as powerful tools for optimization purposes. Finally, AI and ML are increasingly employed to aid enzyme production and modification (Ge et al., 2023). With the expanding role of AI/ML in enzyme engineering, both network models and algorithms are imperative in the design of lipase for industrial applications.

Traditional physics-based protein design methodologies like Rosetta use iterative search strategies to navigate protein space through modeling the biophysical interactions governing the designed sequences, which is computationally expensive and prone to stalling at local minima, hindering the discovery of diverse functional variants (Woolfson, 2021). Future efforts should focus on lipase engineering aided with AI/ML. Deep learning models for structure-to-sequence protein design using graph neural networks (Dauparas et al., 2022, 2025; Yang et al., 2023) have become powerful tools for suggesting diverse functional sequences and for generating designs with improved expression, thermostability, and function (Sumida et al., 2024). More ambitiously, generative artificial intelligence models can design de novo proteins (Winnifrieth et al., 2024) with desirable lipase functions and properties. While early models were mostly validated on binding (Alamdari et al., 2023; Watson et al., 2023), a new generation of models promises to design functional de novo enzymes (Abern et al., 2025). This further expands the diversity of biocatalysts for novel applications.

However, it is difficult to meet industrial requirements with a single enzyme, which prompts the emergence of cascade biocatalysis. Besides the common physical and chemical methods (like the immobilization method) to couple different enzymes to increase catalytic performance, protein engineering is an alternative pathway to achieve that. Fusion proteins, achieved by introducing a linker between enzymes, facilitate

the transportation of substrates among active sites. Feng et al. (2021) co-expressed the ester synthesis enzymes alcohol acetyltransferase 1 and butanol dehydrogenases A with parallel heterodimeric coiled coils named CC-Di-A/B to shorten the catalytic distance. Protein-based encapsulation can be achieved through the engineering of virus or bacterial microcompartments (BMC), making this system a viable carrier for cascade biocatalysis. Wagner et al. (2017) demonstrated that β -galactosidase, Est5 esterase, and glycerol dehydrogenase can be relocated to the BMC and isolated in the microcompartment shell while preserving their enzymatic functions. Moreover, compartmentalization based on liquid-liquid phase separation, which enables the co-localization of sequential enzymes within confined microenvironments, can also be applied to lipase-mediated catalytic reactions to enhance catalytic efficiency. With the advancements in sequencing and high-throughput analytical technologies, multi-omics tools have garnered increasing attention for their ability to help scientists unravel host metabolism and gene regulation. Together with AI/ML, these tools can not only enable the optimization of lipase production but also open the door to one-pot reactions in chassis cells, streamlining the entire process from lipase expression to target product biosynthesis.

CRediT authorship contribution statement

Geng Wang: Writing – original draft, Writing – review & editing, Visualization. **Asma Abdella:** Writing – original draft. **Mohamadali Fakhari:** Writing – original draft. **Jie Dong:** Writing – review & editing. **Kevin K. Yang:** Writing – review & editing. **Shang-Tian Yang:** Conceptualization, Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare no competing financial interest.

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